In [15]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classificatior
from sklearn.preprocessing import LabelEncoder
from sklearn import svm

sns.set_theme(style="whitegrid")
```

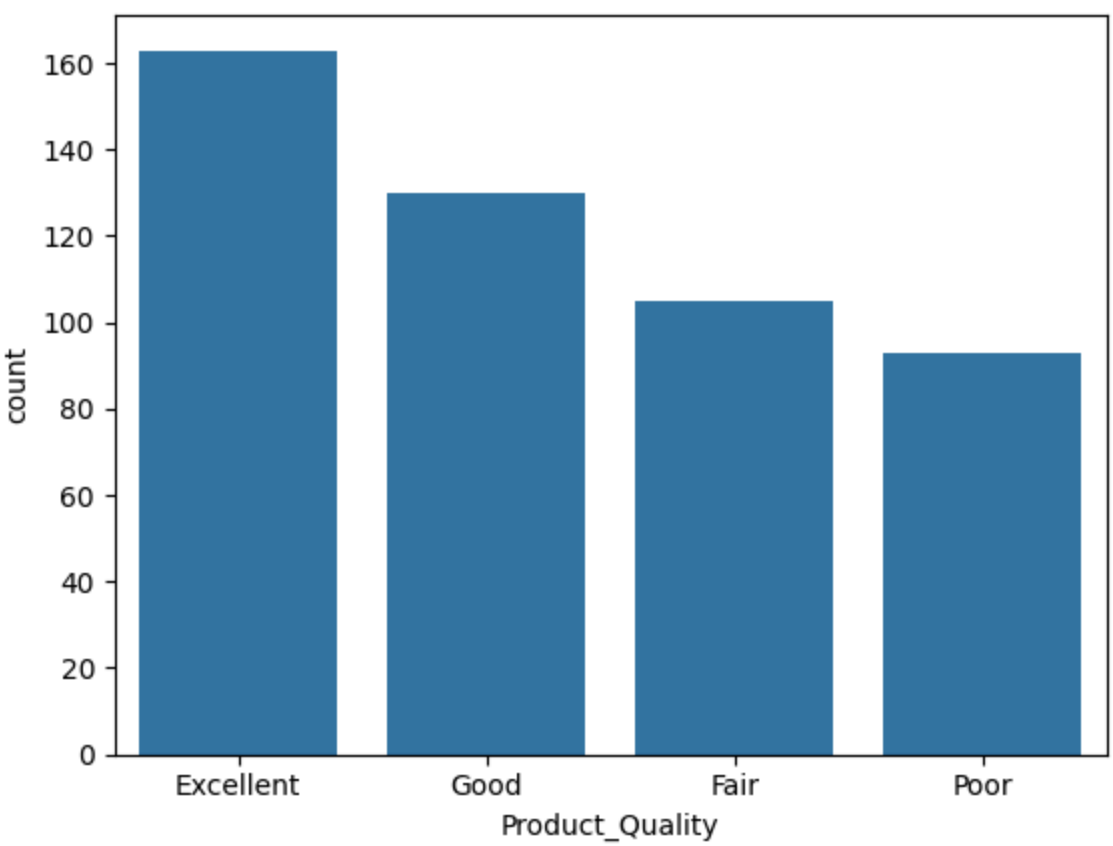
In [16]:
```
df = pd.read_excel ('data/mobile-info.xlsx') #for an earlier version of Exce
```

In [17]:
```
df.head(4)
```

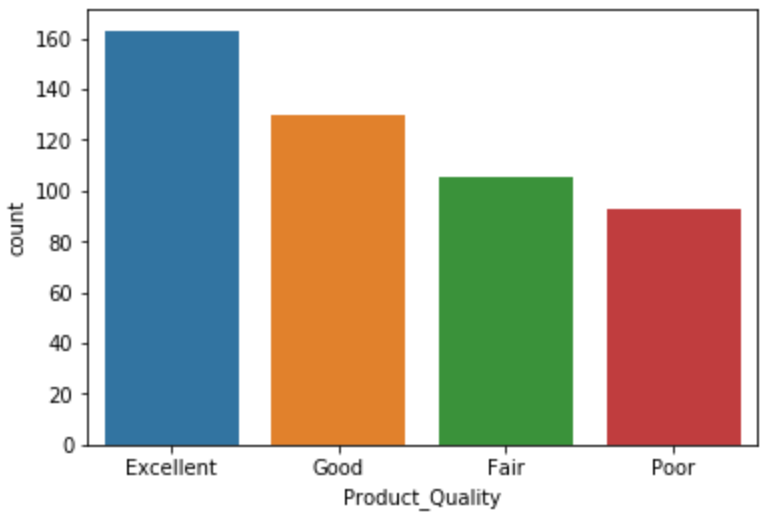
Out[17]:

	Product_Name	Seller_Name	Overall_Rating	Overall_Review	Seller_Rating	A
0	RedmiNote6Pro	FlashstarCommerce	352708	32857	4.7	
1	RedmiNote6Pro	SuperComNet	352708	32857	4.3	
2	RedmiNote7	SuperComNet	10387	1057	4.3	
3	RedmiY1	samaymobiles	17421	1634	3.1	

In [5]:
```
sns.countplot(x=df['Product_Quality']) # Added x= for clarity
plt.show()
```



In [33]:
```
sns.countplot(df['Product_Quality'],label="Count")
plt.savefig('Product Quality')
plt.show()
```



In [8]:
```
lab_enc = LabelEncoder()
```

In [11]:
```
# Encoding specific columns
df['Product_Name'] = lab_enc.fit_transform(df['Product_Name'])
df['Seller_Name'] = lab_enc.fit_transform(df['Seller_Name'])
df['Product_Quality'] = lab_enc.fit_transform(df['Product_Quality'])
```

In [13]:
```
print(df.columns)
```

```
Index(['Product_Name', 'Seller_Name', 'Overall_Rating', 'Overall_Review',
       'Seller_Rating', 'Average_Rating', 'Product_Quality'],
      dtype='str')
```

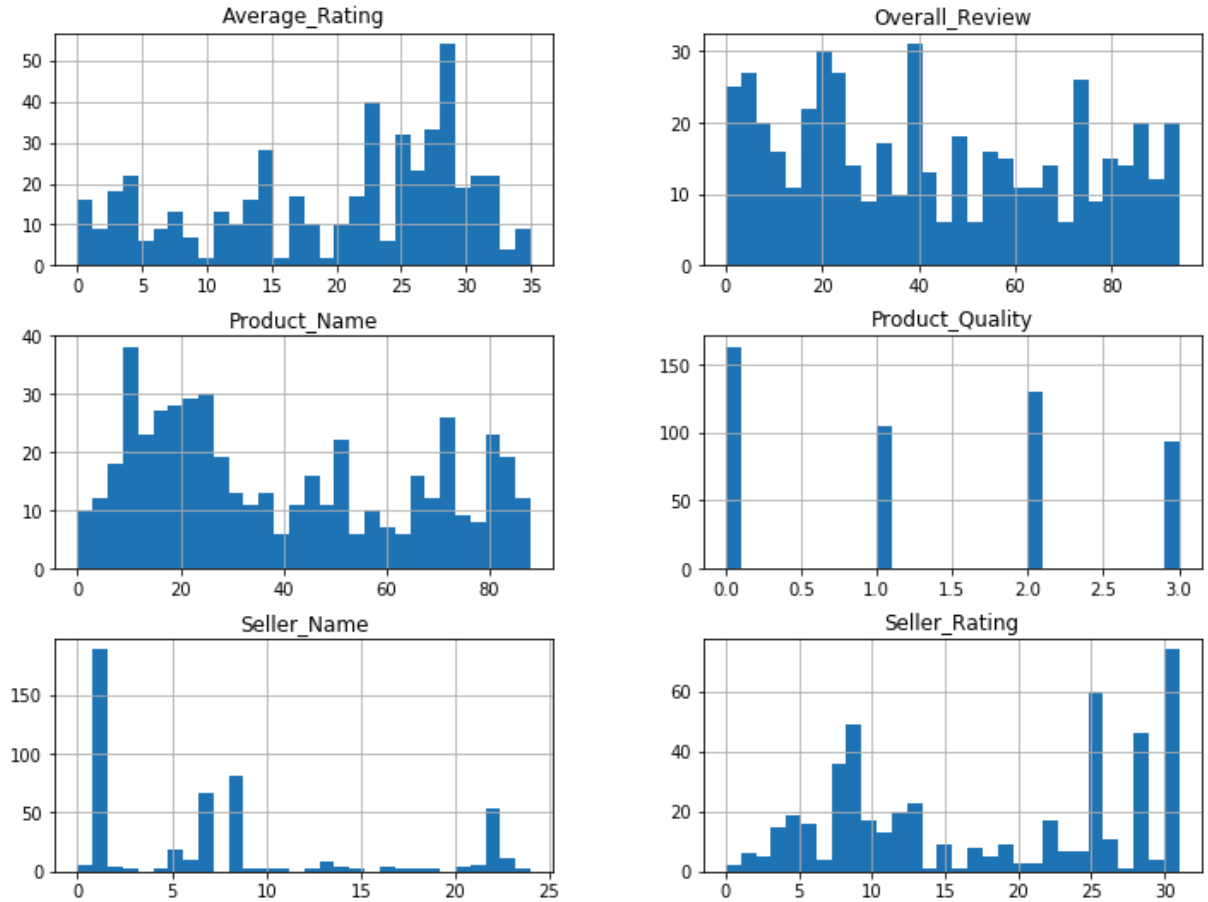
In [9]:
```
feature_cols = ['Seller_Name', 'Seller_Rating', 'Overall_Review', 'Overall_F
X = df[feature_cols] # Features
y = df.Product_Quality # Target variable
```

In [10]:
```
df.head(2)
```

Out[10]:

	Product_Name	Seller_Name	Overall_Rating	Overall_Review	Seller_Rating	Averag
0	11	16	96	87	29	
1	77	7	96	87	25	

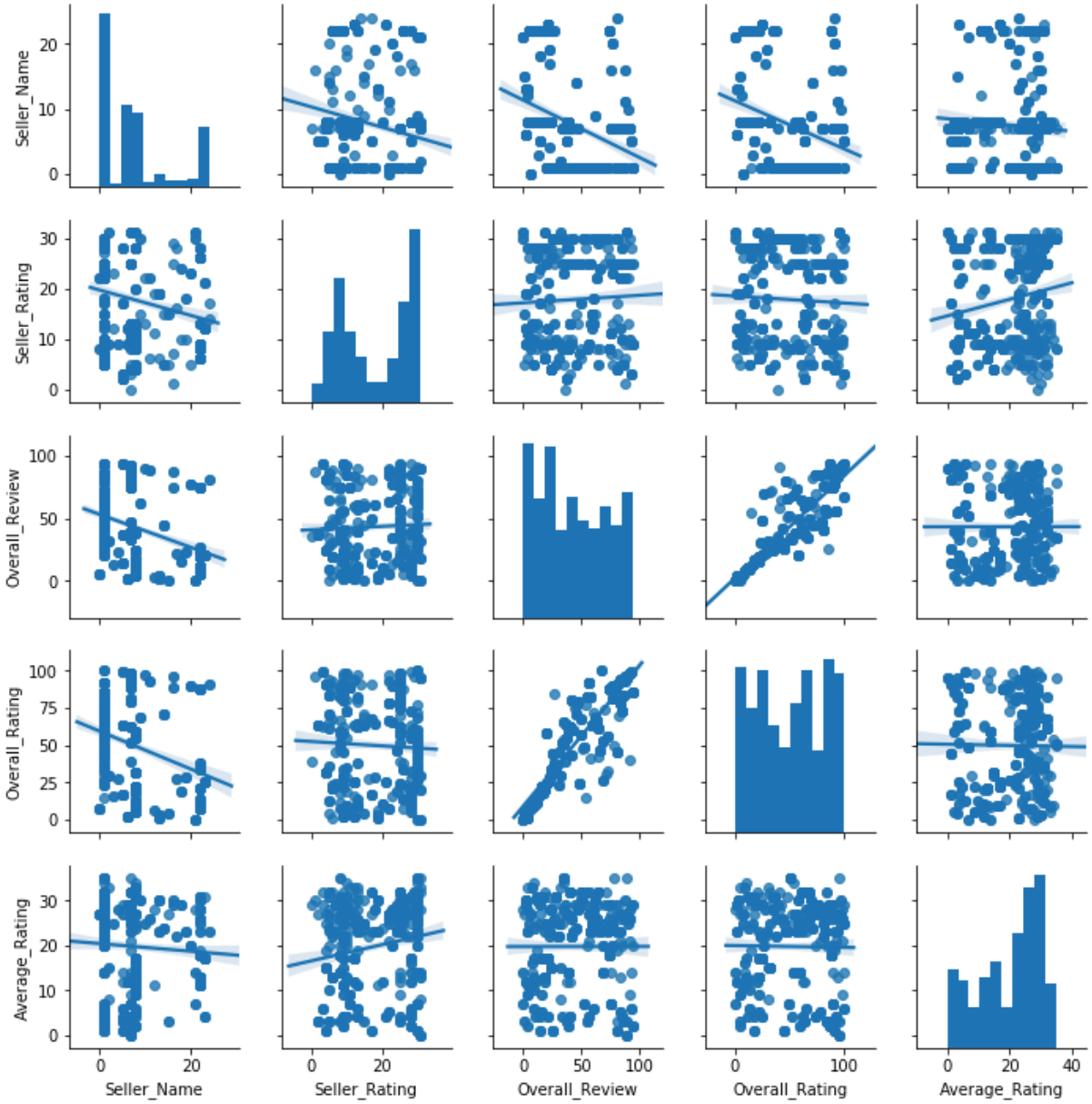
In [11]:
```
import pylab as pl
df.drop('Overall_Rating' ,axis=1).hist(bins=30, figsize=(12,9))
plt.savefig('M_Review')
plt.show()
```



In [12]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [13]:
```
sns.pairplot(X,kind='reg',height=2)
plt.show()
plt.savefig('Mobile_correlation among different attributes')
```

```
C:\ProgramData\Anaconda3\lib\site-packages\scipy\stats\stats.py:1713: Future
Warning: Using a non-tuple sequence for multidimensional indexing is depreca
ted; use `arr[tuple(seq)]` instead of `arr[seq]`. In the future this will be
interpreted as an array index, `arr[np.array(seq)]`, which will result eithe
r in an error or a different result.
  return np.add.reduce(sorted[indexer] * weights, axis=axis) / sumval
```




```
<Figure size 432x288 with 0 Axes>
```

In [14]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [15]:
```
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier

# Create Decision Tree classifer object
clf = DecisionTreeClassifier()

# Train Decision Tree Classifer
clf = clf.fit(X_train,y_train)

#Predict the response for test dataset
y_pred = clf.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
```

```
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 1.0
confusion matrix
[[49  0  0  0]
 [ 0 36  0  0]
 [ 0  0 39  0]
 [ 0  0  0 24]]
             precision    recall  f1-score   support

          0       1.00      1.00      1.00        49
          1       1.00      1.00      1.00        36
          2       1.00      1.00      1.00        39
          3       1.00      1.00      1.00        24

avg / total       1.00      1.00      1.00       148
```

In [16]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [17]:
```
from sklearn.ensemble import RandomForestClassifier

#Create a Gaussian Classifier
clf=RandomForestClassifier(n_estimators=100)

#Train the model using the training sets y_pred=clf.predict(X_test)
clf.fit(X_train,y_train)

y_pred = clf.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
C:\ProgramData\Anaconda3\lib\site-packages\sklearn\ensemble\weight_boosting.
py:29: DeprecationWarning: numpy.core.umath_tests is an internal NumPy modul
e and should not be imported. It will be removed in a future NumPy release.
  from numpy.core.umath_tests import inner1d
Accuracy: 0.9932432432432432
confusion matrix
[[49  0  0  0]
 [ 0 36  0  0]
 [ 0  1 38  0]
 [ 0  0  0 24]]
             precision    recall  f1-score   support

          0       1.00      1.00      1.00        49
          1       0.97      1.00      0.99        36
          2       1.00      0.97      0.99        39
          3       1.00      1.00      1.00        24

avg / total       0.99      0.99      0.99       148
```

In [18]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [19]:
```
from sklearn import svm

#Create a svm Classifier
clf = svm.SVC(kernel='rbf' ,gamma='auto') # Linear Kernel

#Train the model using the training sets
clf.fit(X_train, y_train)

#Predict the response for test dataset
y_pred = clf.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 0.8513513513513513
confusion matrix
[[48  0  1  0]
 [ 5 31  0  0]
 [16  0 23  0]
 [ 0  0  0 24]]
             precision    recall  f1-score   support

          0       0.70      0.98      0.81        49
          1       1.00      0.86      0.93        36
          2       0.96      0.59      0.73        39
          3       1.00      1.00      1.00        24

avg / total       0.89      0.85      0.85       148
```

In [20]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [21]:
```
from sklearn.naive_bayes import GaussianNB

#Create a Gaussian Classifier
gnb = GaussianNB()

#Train the model using the training sets
gnb.fit(X_train, y_train)

#Predict the response for test dataset
y_pred = gnb.predict(X_test)
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 0.9662162162162162
confusion matrix
[[43  0  2  0]
 [ 0 38  0  1]
 [ 2  0 32  0]
 [ 0  0  0 30]]
             precision    recall  f1-score   support

          0       0.96      0.96      0.96        45
          1       1.00      0.97      0.99        39
          2       0.94      0.94      0.94        34
          3       0.97      1.00      0.98        30

avg / total       0.97      0.97      0.97       148
```

In [22]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, ra
```

In [23]:
```
#import the KNeighborsClassifier class from sklearn
from sklearn.neighbors import KNeighborsClassifier
#import metric model to check the accuracy
from sklearn import metrics
#Try running from k=1 through 5 and record testing accuracy
k_range = range(1,5)

score = {}

score_list = []
for k in k_range:
        knn = KNeighborsClassifier(n_neighbors=k)
        knn.fit(X_train,y_train)
        y_pred=knn.predict(X_test)
        score[k] = metrics.accuracy_score(y_test,y_pred)
        score_list.append(metrics.accuracy_score(y_test,y_pred))
# Model Accuracy: how often is the classifier correct?
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
results=confusion_matrix(y_test,y_pred)
print('confusion matrix')
print(results)
print (classification_report(y_test,y_pred))
```

```
Accuracy: 0.8243243243243243
confusion matrix
[[36  0  9  0]
 [ 0 37  1  1]
 [10  2 22  0]
 [ 0  3  0 27]]
             precision    recall  f1-score   support

          0       0.78      0.80      0.79        45
          1       0.88      0.95      0.91        39
          2       0.69      0.65      0.67        34
          3       0.96      0.90      0.93        30

avg / total       0.82      0.82      0.82       148
```

In [ ]: